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Talking Engagement Into Being: A Three-Wave Panel Study Linking Boundary Management Preferences, Work Communication on Social Media, and Employee Engagement

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Widespread use of social media across work and non-work boundaries has heightened concerns about employee engagement in the contemporary workforce. This study examines how employees' boundary management preferences influence their work communication on social media, and how these factors impact their engagement. Results from three waves of survey data (N = 361) demonstrate that work communication mediates the relationship between employee boundary preferences and engagement, supporting the hypothesized causal structure over alternative models. Overall, the findings contribute a novel perspective on employee engagement by showing that mediated work communication plays a central role in constructing engagement, rather than merely demonstrating it. We discuss how organizations can leverage this knowledge to address critical concerns about workplace (dis)engagement in the digital age.

Keywords: Social Media, Technology Use, Employee Engagement, Boundary Management, Work Communication.

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Along with the rise of publicly-available social media (e.g., Facebook and Twitter) and their penetration across private, professional, and public domains, communication scholars and organizational practitioners have devoted increasing attention to the workplace practices and attendant impacts associated with these technologies (boyd & Ellison, 2007; Leonardi, Huysman, & Steinfeld, 2013). A particularly prevalent concern in the context of social media use in organizations is the multivalent involvement of these platforms in employees' work and non-work lives (Fieseler, Meckel, & Ranzini, 2015). Indeed, today's employees frequently use the same social platforms to communicate for and about their work, and to share personal content with friends and family members (Ollier-Malaterre & Rothbard, 2015). The potential for 24/7 connectivity also enables employees to access these platforms both in and outside the workplace, on and off the clock, and through company-issued and personally-owned devices

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(Stohl, Etter, Banghart, & Woo, 2017). In turn, the heightened fluidity and increasing visibility of communication that crosses multiple life domains have complicated the ability for employees to maintain discrete personal/professional boundaries, positioning active boundary management as a critical employment competency of the digital age (Banghart, Etter, & Stohl, 2018; van Zoonen, Verhoeven, & Vliegenthart, 2016).

Concomitantly, broader societal shifts toward knowledge work in the information economy have been accompanied by growing interest in employee engagement—i.e., the degree to which organization members experience meaningful connections to their work, express enthusiasm about their job roles, and maintain commitments to high quality performance (Bakker, Albrecht, & Leiter, 2011). Engagement has become a predominant buzzword in both scholarly and mainstream management circles, the focus of numerous professional associations and consulting initiatives, and a topic of widespread debate—particularly as industry surveys over the last decade routinely indicate low percentages of reported engagement among global employees (Crabtree, 2018). As one recent example, a report from the Gallup organization positions employee engagement as a global crisis based on the research firm's finding that merely 13% of employees are engaged with their work, worldwide (Mann & Harter, 2016).

Discussions around social media dynamics in the workplace offer mixed interpretations around the potential role of these media in facilitating or impeding employee engagement. On the one hand, research has shown that social media use in organizations can enhance job performance (Cao, Guo, Vogel, & Zhang, 2016) and increase employee morale and feelings of cultural belonging (Leidner, Koch, & Gonzalez, 2010). On the other, some studies suggest that social media use in the workplace is counterproductive (Andreassen, Torsheim, & Pallesen, 2014) and generates a host of boundary management issues and tensions for employees (Pike, Bateman, & Butler, 2013). While management voices concern about employees "cyberloafing" and "cyberslacking" (i.e., wasting time and losing productivity by using social media for personal matters while at work; Andreassen et al., 2014), employees contemplate issues such as whether to accept friend requests from bosses and coworkers, and how to manage personal information and conversations that spill over from public online environments into the workplace (Peluchette, Karl, & Fertig, 2013). Yet, even as recent communication scholarship specifies a link between social media use and various forms of engagement (e.g., interpersonal, political, civic, etc.; Dijkmans, Kerkhof, Buyukcan-Tetik, & Beukeboom, 2015; Lovejoy & Saxton, 2012), the relationship between mediated work communication and employee engagement remains largely unexamined and undertheorized.

This study examines how employees' preferences for managing boundaries (i.e., integrating vs. segmenting work/non-work elements online; Ollier-Malaterre & Rothbard, 2015) influence their work communication on social media, and how these factors impact their engagement. We define *work communication on social media* as those communicative acts in which employees share information about their job tasks, organizations, occupations, and/or industries through publicly-available platforms (i.e., Facebook and Twitter rather than enterprise platforms such as Yammer or Slack; Leonardi et al., 2013; van Zoonen et al., 2016), typically through personal and individually-owned accounts. Examining the links between employees' boundary preferences, work communication on social media, and engagement holds both theoretical and practical importance. Theoretically, our longitudinal investigation unpacks the role of mediated work communication in relation to boundary preferences and employee engagement, and in doing so, contributes a novel view of the communication–engagement relationship. Pragmatically, establishing causal priority in these relations helps to explicate some of the circumstances under which social media use may facilitate engagement for some employees, while diminishing engagement for others. We discuss how a clearer understanding of these relations

can help organizations to better manage social media dynamics in the workplace and address critical concerns around employee (dis)engagement.

Theoretical background and hypotheses

This research is informed by work/non-work boundary theory (Ashforth, Kreiner, & Fugate, 2000; Nippert-Eng, 1996) and recent theorizing linking social media use to various forms of engagement. The following sections explicate these theoretical connections and present a set of hypotheses regarding the anticipated links among employees' preferences for managing work/non-work boundaries, the frequency of their work communication on social media, and their engagement with work.

Boundary preferences and work communication on social media

Boundaries are "physical, temporal, and cognitive limits that define domains as separate from one another and define components within domains" (Kreiner, Hollensbe, & Sheep, 2006, p. 1319). According to boundary theory (Nippert-Eng, 1996), people construct boundaries to simplify their environments and differentiate domains of life experience. To that end, individuals develop unique preferences for separating or integrating work/non-work elements, which in turn influence the strategies they use to manage conflicting demands between domains. Whereas segmentation preferences suggest a desire to maintain rigid and impermeable boundaries to keep elements of work and non-work domains separate (e.g., imposing restrictions on mediated work communication during off-hours and/or outside of the workplace; Olson-Buchanan & Boswell, 2006), integration preferences are typically associated with more flexible and permeable boundaries that enable people to blend elements from different social domains together (e.g., using personal devices to address work-related matters while at home; Bulger, Matthews, & Hoffman, 2007).

Given the omnipresence of new media in both social and organizational life, employees develop rules and preferences for communicating on these platforms across their personal/non-work, professional/work, and public/political roles (Ollier-Malaterre & Rothbard, 2015). Indeed, social media are platforms through which employees enact boundary rigidity and permeability, and recent scholarship has linked employees' boundary preferences to the choices they make around those with whom they connect, how they structure their social ties, and the kinds of information they share in mediated communication environments (Marwick & boyd, 2011; van Zoonen et al., 2016). Building on this work, we argue that the frequency with which employees communicate for and about their work on social media is directly related to their integration or segmentation preferences. In particular, we expect that individuals with integration preferences may be more inclined to engage in work communication through their personal social media accounts, thereby allowing their work and non-work lives to blend together online. At the same time, it is likely that employees with segmentation preferences may be less inclined to engage in work communication on personal social media accounts, as this would involve combining work/non-work identities, information, and audiences in an integrated online environment.

H1: Employees' preferences for integrating work/non-work domains will be associated with increases in work communication on social media.

Work communication on social media and employee engagement

Across management literature, scholars typically conceptualize engagement as an employee trait (i.e., a stable characteristic associated with positive views of life and work) or a psychological state (i.e., a cognitive condition involving heightened energy, commitment, resilience, and effort at work), while

paying minimal attention to the underlying communicative dynamics that shape engagement in the first place (Macey & Schneider, 2008; Schaufeli & Bakker, 2004). To wit, the prevailing assumption is that trait and/or state engagement precede desirable work behaviors such that increases in engagement lead to more positive employee and workplace outcomes. For example, when viewing engagement as a pre-existing cognitive state or trait, communication is typically conceptualized as a discretionary effort or contextual performance by employees (Macey & Schneider, 2008), and social media are treated as instruments through which employees can *express* enthusiasm or *demonstrate* their positive affect toward work. Yet, there is limited empirical support for such causal prioritization.

Missing from the literature is a perspective that acknowledges how employees construct and enact engagement through communicating for and about their work on social media—a constructive view of the communication–engagement relationship (Fairhurst & Putnam, 2004, 2014). Social media not only facilitate information-sharing *about* employees' work engagement (i.e., with work-related communication representing engagement) but can also serve as a *model* for generating engagement in the first place (i.e., sharing work-related messages serves to communicatively construct engagement). Contemporary scholars of engagement also specify that employees are not simply passive actors who let work and life happen to them, but rather are active agents who participate in constructing their own work environments and engagement (Bakker et al., 2011). Along these lines, Bakker et al. (2011) propose that employees may attempt to conserve their engagement through behaviors such as job crafting (e.g., customizing work interactions to align with their goals) and point to the need for more research on the specific tactics employees use to maintain or increase engagement.

From a constructivist communication perspective (Fairhurst & Putnam, 2004, 2014), organizational phenomena such as work-related social media use are not merely products of pre-existing states (e.g., engagement) or perpetually evolving patterns of organizing, but are grounded in action; that is, they are both processes and products of communicative practices. Building from these insights, we propose an alternative view of the communication–engagement relationship—a co-constructive perspective—in which work communication on social media both demonstrates engagement and plays a fundamental role in constructing it. In this conception, engagement is not simply a trait or state that leads employees to communicate about work online; it is also a process wherein communicating for and about work aids employees in talking their engagement into being. To that end, our second hypothesis is:

H2: More frequent work communication on social media will be associated with heightened levels of employee engagement.

Boundary management preferences and employee engagement

As a last step in our model, we propose that the impact of employees' boundary management preferences on their engagement will be mediated, at least in part, by the frequency of their work communication on social media. Existing studies have already demonstrated that boundary management preferences are related to engagement (Kossek, Lautsch, & Eaton, 2006) and that social technology use partially mediates the association between boundary preferences and engagement-relevant outcomes such as psychological detachment at work (Park, Fritz, & Jex, 2011). By extension, it seems likely that employees' preferences for integrating work/non-work domains might facilitate an attachment to work that underpins engagement, with work communication on social media further intensifying that process. Hence, if employees use social media to enact their boundary preferences, work communication on social media may partially mediate the association between employees' preferences for integrating work/non-work domains and their engagement.

H3: Work communication on social media will partially mediate the boundary management preferences and employee engagement relationship.

Methods

Procedure and participants

Participants supplied data through a web-based survey administered by PanelClix¹ at three different times with 2.5 months between the measurement points. At time one (T1), we employed quota sampling to obtain the predetermined number of 1,008 respondents. A total of 7,000 Dutch employees received the questionnaire, but only the first 1,008 respondents were allowed to participate in the study, after which the survey was closed. At time two (T2), 578 of the 1,008 employees completed the questionnaire (a 42.7% drop-out percentage). At time three (T3), 361 of the 578 employees who completed the questionnaire at T1 and T2 completed the questionnaire (a 37.5% drop-out percentage). Thus, 361 employees completed all three waves, meaning that 647 of the 1,008 respondents from T1 did not participate at T2 or T3.

Responses to standard demographic questions indicated that the majority of the participants (56%) were male. The participants were on average 47 years ($SD = 11.06$). In addition, the average employee in the final sample held 14.89 years ($SD = 11.98$) of organizational tenure and reported an average work week of 36.72 hours ($SD = 8.56$). We also examined selective drop-outs through the comparison of the scores of participants who dropped out ($N = 647$) to the scores of those who completed all three waves (i.e., the final sample, $N = 361$). This indicated that men were slightly overrepresented in the final sample (56% of participants were male compared to 45% in the initial sample; $\chi^2 = 12.23$, $p > .001$) and that the average age was slightly higher among the final sample ($M = 47.00$, $SD = 11.06$) compared to that of the initial sample at T1 ($M = 42.95$, $SD = 11.52$; $t = -5.42$, $p < .001$).

We assessed causal homogeneity between the drop-out sample and the final sample through cross-sectional multi-sample (i.e., drop-out versus survivor) structural equation analysis. In doing so, we examined whether the causal relationships were different for the drop-out group in comparison to the survivor group in the final sample. The findings demonstrate that the effect of boundary preference on work communication on social media was somewhat larger in the final sample [$b^* = .231$, BC95% (.161; .303) $p = .001$] compared to the initial sample including participants who dropped out [$b^* = .147$, BC95% (.093; .202) $p = .001$]. However, this difference was not significant ($z = 1.702$, $p = .088$). There were also no differences in the effects between work communication on social media and engagement; hence, it is unlikely that different causal dynamics influenced participants to drop out of the study. Finally, the Missing Completely at Random test indicated no systematic pattern of missing values ($\chi^2 = 106.96$, $df = 94$, $p = .170$).

Measures

First we measured *boundary preferences* along a continuum based on existing measures by Kreiner, Hollensbe, and Sheep (2006) and Ilies, Wilson, and Wagner (2009). Participants rated their agreement with six statements relating to integration and segmentation preferences on a 5-point scale ranging from: 1 = not at all to 5 = very much. These items were used to create three parcels that reflect the continuum ranging from integration preferences to segmentation preferences (Rothbard, Phillips, & Dumas, 2005). Factor loadings ranged from .90 to .96 at T1; .92 to .95 at T2; and .92 to .96 at T3.

Second we measured *work communication on social media* with five items derived from an existing instrument by van Zoonen et al. (2016). The measure assesses the frequency of sharing work-related messages through publicly-available social media. In particular, we asked respondents to

indicate the frequency with which they used social media in the past week to share information about: (a) work projects, (b) their organizations, (c) their organization's products or services, (d) their industry; and (e) their daily work activities. Participants responded to a Likert-type scale ranging from: 1 = never to 5 = very often (multiple times a day) for each item. Factor loadings were between .74 and .92 at T1; .76 and .92 at T2; and .71 and .89 at T3.

Finally, we measured *engagement* using the items representing vigor in the Utrecht Work Engagement Scale (UWES; Schaufeli & Bakker, 2004). The measure asks participants to report how frequently they experience certain conditions of engagement (e.g., "When I get up in the morning, I look forward to starting the workday"). Responses were documented on a Likert-type scale ranging from: 0 = never to 6 = always (daily). Factor loadings ranged from .76 to .94 at T1; .73 to .95 at T2; and .70 to .94 at T3.

Data analysis

In this study we examine mediation involving two causal relations (i.e., boundary preferences on work communication on social media, and work communication on social media on employee engagement). To establish causality, one variable must precede the other in time. Thus, we focus on traditional regression-based analysis, particularly on linear relationships that allow the examination of individual differences across time points (Cole & Maxwell, 2003). Curve estimations show that the relationships between boundary preferences, work communication on social media, and employee engagement are linear. Hence, covariance-based structural modeling in statistical software package Amos can be applied. We examined the mediational effects using structural equation modeling by following the five-step process outlined by Cole and Maxwell (2003). Steps 1 and 2 involve tests for the robustness of the measurement model such as establishing longitudinal factorial invariance. Steps 3 and 4 involve testing competing and progressively constrained models. Finally, in step 5 mediational parameters are examined.

Competing models include a baseline model ($M_{(\text{baseline})}$)—that examines the extent to which variables in the model remain stable over time—by assessing the autoregressive paths—i.e., the regression weights between the same variables across waves (e.g., boundary preferences at T1 and boundary preferences at T2). We evaluated the baseline model by testing it against the causal model $M_{(\text{causal})}$, the reversed causal model $M_{(\text{reversed})}$ and the reciprocal model $M_{(\text{reciprocal})}$. Note that $M_{(\text{causal})}$ extends the $M_{(\text{baseline})}$ by examining the effects of boundary preferences on work communication on social media and of work communication on social media on employee engagement *between* time points. Thus, the causal model reflects the assumption that employees with integration preferences share more work-related information than those with segmentation preferences (H1), and that work communication on social media is positively related to employee engagement (H2). In contrast, the reversed causation model $M_{(\text{reversed})}$ includes effects in the opposite direction from engagement to work communication on social media and from work communication on social media to boundary preferences. The reciprocal model $M_{(\text{reciprocal})}$ includes the reciprocal relationships between boundary preferences, work communication on social media, and engagement (i.e., including all effects of the $M_{(\text{causal})}$ and $M_{(\text{reversed})}$). We compared different models through a $\Delta\chi^2$ test. Additionally, we used multiple fit indices to examine model fit—i.e., TLI; CFI; SRMR; and RMSEA.

Results

Measurement model

The first step is estimating the measurement model. At all three waves of data collection, the measurement model including employees' preferences for managing boundaries, work communication on

social media, and employee engagement shows satisfactory model fit: $\chi^2 (666) = 1,931.25$; TLI = .91; CFI = .92; RMSEA = .073 (CI: .069, .076), and SRMR = .04. Factor correlations within waves ranged from .08 to .36 demonstrating that the latent constructs are sufficiently distinct (i.e., discriminant validity). Correlations across waves between the same factors ranged from .55 to .80. This suggests that the constructs are relatively stable across measurement points (see Table 1).

Next, we examined factor loadings and explained variances to assess convergent validity. The observed variables yielded significant and sizable standardized factor loadings between .70 to .96. Each of the observed variables had an explained variance between .49 and .93 on the respective latent construct. Additionally, the alpha coefficients reported in Table 1 are all above .90. Hence, the measurement model adequately measures boundary preferences, work communication on social media, and employee engagement.

Longitudinal factorial invariance

The second phase involves comparing various parameters and examining their equivalence across waves. Examining longitudinal factorial invariance is important as this helps to provide evidence for the imperative assumption that the fundamental meaning of the latent variables is consistent across measurement points. We examined factorial invariance by modeling progressively constrained models and comparing all models to more restricted (nested) models.

Longitudinal factorial invariance was established as the structural models were invariant over time: (1) configural invariance [$\Delta\chi^2 (20) = 20.06, p = .454$], (2) loading invariance [$\Delta\chi^2 (46) = 38.42, p = .778$], (3) intercept invariance [$\Delta\chi^2 (58) = 52.73, p = .787$], and (4) residual invariance [$\Delta\chi^2 (84) = 99.18, p = .123$]. These results demonstrate that the meanings of the observed variables and latent constructs in the model were invariant. In other words, they did not change over time. Furthermore, the invariance tests show that the causal parameters met the stationarity assumptions (Cole & Maxwell, 2003).

Structural regression models

The next steps refer to testing the hypotheses by estimating several competing models (see Table 2 for model fit statistics). Generally, every estimated regression model demonstrated acceptable model fit as the threshold values for the model fit indices were met. In what follows, we first discuss the comparisons used to determine which model best fit the data before presenting the results of our hypothesis testing.

The causal model (M_{causal}) showed better fit than the baseline model (M_{baseline}) $\Delta\chi^2 (4) = 28.85, p < .001$. Hence, adding the cross-lagged effects of boundary preferences to work communication on social media, and of work communication on social media to employee engagement made a substantial difference. Successively, we estimated the reverse causation model (M_{reversed}). The reversed model contains the effects in the opposite direction—i.e., from employee engagement to work communication on social media, and from work communication on social media to boundary preferences. The findings demonstrate that model fit did not improve in comparison to the model with only autoregressive effects (M_{baseline} : $\Delta\chi^2 (4) = 3.96, p = .411$). Finally, we estimated the reciprocal model ($M_{\text{reciprocal}}$), including cross-lagged effects in both causal directions. Hence, the reciprocal model includes the effects of boundary preferences on work communication on social media and vice versa, and the effects between work communication on social media and employee engagement and vice versa. The fit of the reciprocal model improved significantly in comparison to that of the baseline model [M_{baseline} : $\Delta\chi^2 (8) = 33.58, p < .001$] and the reversed model [M_{reversed} : $\Delta\chi^2 (4) = 29.62, p < .001$]. Compared to the more parsimonious causal model, the reciprocal model did not fit better [M_{causal} : $\Delta\chi^2 (4) = 4.73, p = .316$], hence the causal model is preferred, despite nearly equal model fit statistics.

Table 1 Correlations and Descriptive Statistics

Constructs	Mean (SD)	1	2	3	4	5	6	7	8	9	10	11	12
<i>Time 1</i>													
1. Boundary preference T1	1.40 (1.42)	.95											
2. Social media use T1	1.54 (.83)	.30*	.93										
3. Engagement T1	5.24 (1.19)	.22*	.18*	.93									
<i>Time 2</i>													
4. Boundary preference T2	1.38 (1.40)	.80*	.29*	.19*	.96								
5. Social media use T2	1.50 (.81)	.31*	.56*	.10	.32*	.93							
6. Engagement T2	5.15 (1.21)	.21*	.18*	.79*	.21*	.11*	.93						
<i>Time 3</i>													
7. Boundary preference T3	1.38 (1.42)	.79*	.29*	.13*	.77*	.28*	.14*	.96					
8. Social media use T3	1.48 (.74)	.33*	.55*	.08	.30*	.62*	.08	.36*	.91				
9. Engagement T3	5.13 (1.21)	.23*	.18*	.78*	.22*	.13*	.80*	.17*	.15*	.92			
<i>Controls at Time 1</i>													
10. Gender	1.44 (.50)	.01	−.07	.02	.06	.00	−.06	.03	−.01	−.06	–		
11. Age	47.00 (11.06)	.04	−.01	.14*	.05	−.01	.16*	−.01	−.07	.16*	−.20*	–	
12. Working hours p/w	36.72 (8.56)	.13	.10	.16*	.21*	.16*	.18*	.14*	.08	.16*	−.36*	.01	–
13. Tenure	14.89 (11.98)	.00	−.12*	.06	−.02	−.06	.14*	−.05	−.12*	.10	−.19*	.52*	−.08

Note: Values on the diagonal are alpha coefficients. Values in italics represent stability coefficients of the same constructs across waves. Significant correlations are flagged *.

Table 2 Model Fit Statistics

Name	Model	χ^2	<i>df</i>	TLI	CFI	RSMEA (95% CI)	SRMR	$\Delta \chi^2$ (Δdf)	Model comparison
M _m	Measurement model	1,931.25	666	.91	.92	.073 (.069; .076)	.04		
M _{baseline}	Only autoregressive structural paths	2,040.66	688	.91	.91	.074 (.070; .078)	.06		
M _{causal}	M _{baseline} + BP > WCSM > Engagement	2,011.81	684	.91	.91	.073 (.070; .077)	.06	28.85** (4)	M _{baseline} vs M _{causal}
M _{reversed}	M _{baseline} + Engagement > WCSM > BP	2,036.70	684	.91	.91	.074 (.070; .078)	.08	3.96 <i>n.s.</i> (4)	M _{baseline} vs M _{reversed}
M _{reciprocal}	M _{reversed} + M _{causal}	2,007.08	680	.91	.92	.074 (.070; .077)	.05	33.58** (8)	M _{baseline} vs M _{reciprocal}
								4.73 <i>n.s.</i> (4)	M _{causal} vs M _{reciprocal}
								29.62** (4)	M _{reversed} vs M _{reciprocal}

Note: ** = .05; BP = boundary management preferences; WCSM = work communication on social media.

Hypothesis testing

We examined the structural relationships of the causal model to test our hypotheses. Notably, the autoregressive effects of boundary preferences between Time 1 and Time 2 [$b^* = .853$, BC95% (.795; .920) $p = .001$], and Time 2 and Time 3 [$b^* = .813$, BC95% (.741; .893) $p < .001$] indicated that while participants' boundary preferences were relatively stable over time. In turn, work communication on social media was somewhat less stable across times as indicated by the autoregressive effects from Time 1 on Time 2 [$b^* = .489$, BC95% (.329; .690) $p < .001$] and from Time 2 on Time 3 [$b^* = .422$, BC95% (.254; .612) $p < .001$]. Lastly, the effects of employee engagement at Time 1 on employee engagement at Time 2 [$b^* = .864$, BC95% (.773; .968) $p < .001$] and from employee engagement at Time 2 on employee engagement Time 3 [$b^* = .583$, BC95% (.420; .750) $p < .001$] indicated relative stability.

Our first hypothesis (H1) asserted that employees' boundary preferences would have a positive impact on work communication on social media. The causal model shows that boundary management preferences at Time 1 have a direct effect on work communication on social media at Time 2 [$b^* = .080$, BC95% (.032; .133) $p < .001$]. Similarly, boundary preferences at Time 2 also predict work communication on social media at Time 3 [$b^* = .062$, BC95% (.021; .105) $p = .004$]. There was no significant effect in the opposite direction as work communication on social media at Time 1 did not affect boundary preferences at Time 2 [$b^* = .117$, BC95% (-.011; .259) $p = .075$] nor did work communication on social media at Time 2 affect boundary preferences at Time 3 [$b^* = .054$, BC95% (-.115; .221) $p = .502$].

Hypothesis 2 suggested that increases in the frequency of work communication on social media would be accompanied by increases in employee engagement. The final model indicates that the effect for work communication on social media at Time 1 did not significantly impact employee engagement at Time 2 [$b^* = .070$, BC95% (-.027; .176) $p = .240$]. However, there was a significant positive effect of work communication on social media at Time 2 on employee engagement at Time 3 [$b^* = .133$, BC95% (.033; .250) $p = .010$]. Notably, employee engagement at Time 1 did not significantly influence work communication on social media at Time 2 [$b^* = -.023$, BC95% (-.076; .024) $p = .344$]. Conversely, the effect of employee engagement at Time 2 on work communication at Time 3 [$b^* = -.014$, BC95% (-.054; .026) $p = .432$] was not significant. Thus, these findings offer limited support for H2.

Finally, our third hypothesis (H3) proposed an indirect effect of employees' boundary preferences on employee engagement through work communication on social media. The results support this notion as boundary preferences at Time 1 yielded a significant effect on employee engagement at Time 3 through work communication on social media at Time 2 [$b^* = .011$, BC95% (.003; .026) $p = .007$]. Notably, boundary preferences at Time 1 had a significant effect on employee engagement at Time 3 in a model without mediators [$b^* = .196$, BC95% (.104; .296) $p = .001$]. In the model with mediators this effect was smaller [$b^* = .080$, BC95% (.023; .144) $p = .024$], thus supporting the partial mediation proposed in H3 (see Figure 1).

Discussion

This study examined the extent to which employees' preferences for managing work/non-work boundaries influenced the frequency with which they communicated about work on social media, and the extent to which these factors related to their engagement. Perhaps the most important finding is that, irrespective of individuals' boundary preferences, work communication on social media plays a critical role in *constructing* employee engagement rather than merely representing or demonstrating

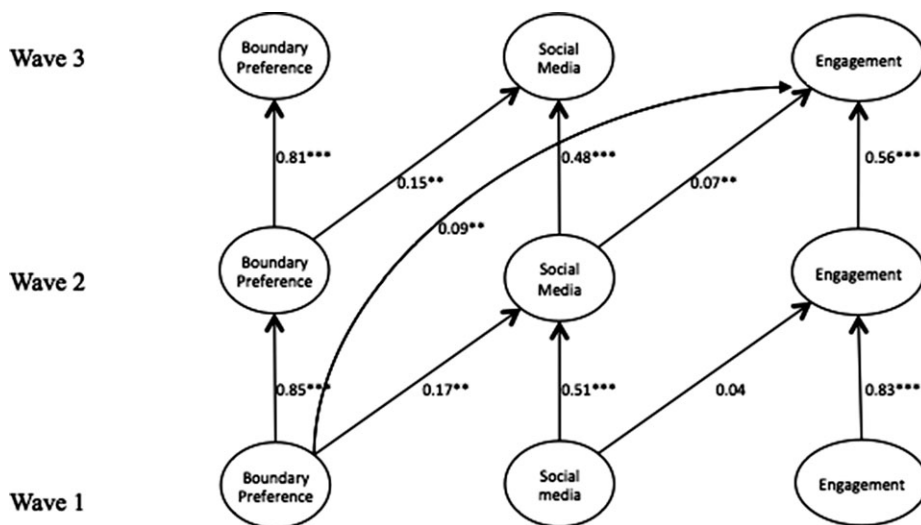


Figure 1 Regression model.

Note: For sake of clarity the figure represents only the regression weights between the latent factors. The observed variables and correlations between latent factors were omitted from this visual representation. Significance is indicated by: *** $p < .001$; ** $p < .05$.

it. As our analysis revealed, employees' preferences for managing boundaries at T1 had a significant effect on the frequency of their mediated work communication at T2 while controlling for reported frequencies of work communication on social media at T1. In turn, work communication on social media at T2 had a significant effect on employee engagement at T3, while accounting for previous levels of engagement. Finally, the relationship between boundary preferences at T1 and employee engagement at T3 was partially mediated by work communication on social media. The findings did not support reverse causality.

Theoretical implications

Although previous studies have specified links between social media communication and engagement-related concepts, this study is the first to foreground computer-mediated communication in the engagement process and to empirically demonstrate a causal relationship between work communication on social media and employee engagement. Our finding that work communication on social media increases employee engagement both complements and extends existing theory. Notably, employee engagement did not reciprocally influence work communication on social media. Thus, beyond supporting the well-established notion that communication constructs individual and organizational realities (Fairhurst & Putnam, 2004, 2014), our findings provide further evidence that work communication on social media serves to shape and actively construct employee engagement. This finding stands in stark contrast to much of the existing literature, which tends to position employee engagement primarily as a personality trait or cognitive state that precedes particular job activities and communicative acts such as sharing work-related information online. Hence, by demonstrating that work communication on social media irrevocably precedes and even aids in constructing employee engagement over time, our study reverses the traditional figure-ground relationship between engagement and everyday work practices.

Our findings also reinforce those of previous studies, which have shown that employees enact boundary preferences in their use of new media (Olson-Buchanan & Boswell, 2006; Park et al., 2011). In turn, we add to this conversation by demonstrating that employees' integration preferences and the frequency of their work communication on social media also have a direct and positive impact on engagement. Conversely, employees with segmentation preferences appear less inclined to utilize social media to share work-related messages and also report lower engagement. Hence, employees' work communication on social media is, at least in part, a reflection of their boundary management preferences and these mediated communicative acts play a critical role in constructing employee engagement.

Although the focus of our analysis is limited to self-reported communicative practices, considering our results in light of recent work drawing from the affordance perspective on social technology use (Treem & Leonardi, 2013) allows for some additional speculation regarding possible explanations behind this relationship. Clearly, the widespread use of social media across work and non-work domains provides employees with quick and relatively easy ways to stay connected to their work. Recent scholarship also demonstrates that the increased accessibility and efficiency of communication afforded by message visibility and association on social media is positively linked to employee engagement (van Zoonen, Verhoeven, & Vliegenthart, 2017). Our findings further substantiate these claims. In particular, it seems likely that work takes a more prominent place in employees' day-to-day lives—both on and offline—when they routinely communicate about their work practices, job roles, organizations, and industries through social media. Moreover, as work and organization-related issues become more visible and accessible to broader audiences in mediated environments, employees may experience their work as more engaging. The potential for online audiences to acknowledge such messages and even provide feedback (e.g., by “retweeting,” “liking,” or commenting on employees' messages) is also likely to reinforce work communication on social media by employees, and particularly those who prefer to integrate life domains. Overall then, our findings suggest that employees' work/non-work boundary preferences and social media use are fruitful starting points from which to examine engagement as communicatively constructed through the interplay of work-related talk and social media affordances in the digital age.

Notably, while our findings offered support for our hypotheses regarding the causal relationships among work communication on social media and employee engagement, the fact that the mean scores on the boundary preference measure were so low is theoretically surprising. In contrast to the prevailing trends in both popular and academic discourse, which generally frame integration as a desirable “new norm” of organizational life in the digital age (Dumas & Sanchez-Burks, 2015), we find that the majority of participants in our sample were inclined to report segmentation rather than integration preferences. Employees with segmentation preferences also reported lower engagement compared to those who preferred to integrate. In a broader sense, this finding indicates that employees with segmentation preferences may be less engaged with work compared to integrators, or at the very least, may have more difficulty achieving the same levels of engagement as integrators do. Given that segmentation may already prompt difficulties for these individuals as they transition between multiple life roles (Ashforth et al., 2000), our findings offer compelling evidence to suggest that organizations may benefit from searching for alternative ways to help these employees communicatively construct engagement.

Practical implications

Pragmatically, the key question facing organizations then, is how to foster opportunities for employees who hold different boundary management preferences to communicatively construct their engagement. For one, it may behoove managers to attend to the ways in which employees do and do not use

social media for work communication as one potential indicator of their boundary preferences. However, irrespective of employees' boundary preferences, our findings suggest that organizations should direct attention to the opportunities available to employees for communicatively constructing their engagement.

Based on our results, employees with integration preferences seem better equipped to enact their boundary preferences and thereby enhance their engagement through work communication on social media. Thus, for integrators, explicit support from management around the use of publicly-available platforms for work communication may suffice in enhancing engagement. However, given that integrators may comprise the minority in many companies, as they did in our sample, organizations also need to consider different avenues to foster engagement for employees with segmentation preferences. It seems possible that segmenters might be more willing to use social technologies for work communication (and thereby become more engaged) if their messages are restricted to an internal organizational audience and remain inaccessible to external (non-work) constituencies. Thus, management might consider implementing enterprise platforms and/or promoting the use of existing internal communication media in which message visibility is limited to work audiences (Leonardi et al., 2013). Enterprise media such as Yammer or Slack may provide segmenters with alternative channels through which they can communicatively construct engagement without blurring personal/professional boundaries.

In addition, recent scholarship has clearly noted that organizations continue to struggle with work communication on social media, with management increasingly imposing prohibiting guidelines that constrain employee communication and behavior on publicly-available platforms to mitigate reputational, legal, and business risks (Stohl et al., 2017). Yet, our findings indicate that such constraining corporate policies might limit potential opportunities for the enactment of employees' boundary management preferences—particularly for integrators—and by extension, limit employee engagement. Given that policies often include broad and ambiguous guidelines that explicitly constrain online employee communication and behavior (Banghart et al., 2018), integrators might become hesitant in terms of how their managers and colleagues will respond to their social media messages. Overall then, we suggest that management strategies and corporate policies should be implemented in ways that neither limit employees' communicative freedoms nor hinder opportunities to construct engagement.

Limitations and future research directions

Our study has several limitations. For one, despite the strengths of our analyses, we cannot rule out the potential influence of confounding factors not included in our models, for instance, individuals' overall social media use, positive affect, organizational identification, corporate social media policies, and organizational and occupational norms. Although cross-lagged autoregressive models account for the influence of a stable third variable, identifying confounding variables should form the focus of future research in this area.

Second, this research focused on the direct effects between work communication on social media and employee engagement. However, other studies on social media and employee well-being suggest that this relationship may be underpinned by different indirect mechanisms (van Zoonen et al., 2016). Thus, future work should examine additional factors that shape the association between work communication on social media and employee engagement as well as the possible role of exhaustion, given its antipodal link to engagement.

Finally, this study showed that employees generally have little desire to integrate their work and non-work lives online and report using social media for work-related communication relatively infrequently, overall. Given that integration is often privileged in both recent academic literature and popular discourse surrounding work/life boundaries (Dumas & Sanchez-Burks, 2015), our results suggest a

need for more work investigating the ways in which employees with segmentation preferences can communicatively construct engagement in the digital age. Nonetheless, we find that both boundary management preferences and work communication on social media have a significant impact on engagement and thus are important factors for future scholars and managerial practitioners to consider in this domain.

Note

1 Dutch-based research company, their panel is ISO26362 certified.

References

- Andreassen, C. S., Torsheim, T., & Pallesen, S. (2014). Predictors of use of social network sites at work: A specific type of cyberloafing. *Journal of Computer-Mediated Communication*, 19(4), 906–921. doi:10.1111/jcc4.12085
- Ashforth, B. E., Kreiner, G. E., & Fugate, M. (2000). All in a day's work: Boundaries and micro role transitions. *Academy of Management Review*, 25(3), 472–491. doi:10.5465/AMR.2000.3363315
- Bakker, A. B., Albrecht, S. L., & Leiter, M. P. (2011). Key questions regarding work engagement. *European Journal of Work and Organizational Psychology*, 20(1), 4–28. doi:10.1080/1359432X.2010.485352
- Banghart, S., Etter, M., & Stohl, C. (2018). Organizational boundary regulation through social media policies. *Management Communication Quarterly* 32(3), 337–373. doi:10.1177/0893318918766405
- boyd, d. m., & Ellison, N. B. (2007). Social network sites: Definition, history, and scholarship. *Journal of Computer-Mediated Communication*, 13(1), 210–230. doi:10.1111/j.1083-6101.2007.00393.x
- Bulger, C. A., Matthews, R. A., & Hoffman, M. E. (2007). Work and personal life boundary management: Boundary strength, work/personal life balance, and the segmentation-integration continuum. *Journal of Occupational Health Psychology*, 12(4), 365–375. doi:10.1037/1076-8998.12.4.365
- Cao, X., Guo, X., Vogel, D., & Zhang, X. (2016). Exploring the influence of social media on employee work performance. *Internet Research*, 26(2), 529–545. doi:10.1109/HICSS.2012.618
- Cole, D. A., & Maxwell, S. E. (2003). Testing mediational models with longitudinal data: Questions and tips in the use of structural equation modeling. *Journal of Abnormal Psychology*, 112(4), 558–577. doi:10.1037/0021-843X.112.4.558
- Crabtree, S. (2018). Global productivity hinges on human capital development, *Gallup Blog*. Retrieved from http://news.gallup.com/opinion/gallup/225752/global-productivity-hinges-human-capital-development.aspx?g_source=BLOG_TGB&g_medium%20=topic&g_campaign=tiles.
- Dijkmans, C., Kerkhof, P., Buyukcan-Tetik, A., & Beukeboom, C. J. (2015). Online conversation and corporate reputation: A two-wave longitudinal study on the effects of exposure to the social media activities of a highly interactive company. *Journal of Computer-Mediated Communication*, 20(6), 632–648. doi:10.1111/jcc4.12132
- Dumas, T. L., & Sanchez-Burks, J. (2015). The professional, the personal, and the ideal worker: Pressures and objectives shaping the boundary between life domains. *Academy of Management Annals*, 9(1), 803–843. doi:10.1080/19416520.2015.1028810
- Fairhurst, G. T., & Putnam, L. L. (2004). Organizations as discursive constructions. *Communication Theory*, 14(1), 5–26. doi:10.1111/j.1468-2885.2004.tb00301.x

- Fairhurst, G. T., & Putnam, L. L. (2014). Organizational discourse analysis. In L. L. Putnam & D. K. Mumby (Eds.), *The SAGE handbook of organizational communication: Advances in theory, research, and methods* (3rd ed., pp. 271–296). Thousand Oaks, CA: SAGE Publications, Inc.
- Fieseler, C., Meckel, M., & Ranzini, G. (2015). Professional personae: How organizational identification shapes online identity in the workplace. *Journal of Computer-Mediated Communication*, 20(2), 153–170. doi:10.1111/jcc4.12103
- Ilies, R., Wilson, K. S., & Wagner, D. T. (2009). The spillover of daily job satisfaction onto employees' family lives: The facilitating role of work-family integration. *Academy of Management Journal*, 52(1), 87–102. doi:10.5465/AMJ.2009.36461938
- Kossek, E. E., Lautsch, B. A., & Eaton, S. C. (2006). Telecommuting, control, and boundary management: Correlates of policy use and practice, job control, and work-family effectiveness. *Journal of Vocational Behavior*, 68(2), 347–367. doi:10.1016/j.jvb.2005.07.002
- Kreiner, G. E., Hollensbe, E. C., & Sheep, M. L. (2006). On the edge of identity: Boundary dynamics at the interface of individual and organizational identities. *Human Relations*, 59(10), 1315–1341. doi:10.1177/0018726706071525
- Leidner, D., Koch, H., & Gonzalez, E. (2010). Assimilating generation Y IT new hires into USAA's workforce: The role of an enterprise 2.0 system. *MIS Quarterly Executive*, 9(4), 229–242.
- Leonardi, P. M., Huysman, M., & Steinfield, C. (2013). Enterprise social media: Definition, history, and prospects for the study of social technologies in organizations. *Journal of Computer-Mediated Communication*, 19(1), 1–19. doi:10.1111/jcc4.12029
- Lovejoy, K., & Saxton, G. D. (2012). Information, community, and action: How nonprofit organizations use social media. *Journal of Computer-Mediated Communication*, 17(3), 337–353. doi:10.1111/j.1083-6101.2012.01576.x
- Macey, W. H., & Schneider, B. (2008). The meaning of employee engagement. *Industrial and Organizational Psychology*, 1(1), 3–30. doi:10.1111/j.1754-9434.2007.0002.x
- Mann, A., & Harter, J. (2016). The worldwide employee engagement crisis. Retrieved from <http://news.gallup.com/businessjournal/188033/worldwide-employee-engagement-crisis.aspx>
- Marwick, A. E., & boyd, d. (2011). I tweet honestly, I tweet passionately: Twitter users, context collapse, and the imagined audience. *New Media & Society*, 13(1), 114–133. doi:10.1177/1461444810365313
- Nippert-Eng, C. E. (1996). *Home and work: Negotiating boundaries through everyday life*. Chicago, IL: University of Chicago Press.
- Ollier-Malaterre, A., & Rothbard, N. P. (2015). Social media or social minefield? Surviving in the new cyberspace era. *Organizational Dynamics*, 44(1), 26–34. doi:10.1016/j.orgdyn.2014.11.004
- Olson-Buchanan, J. B., & Boswell, W. R. (2006). Blurring boundaries: Correlates of integration and segmentation between work and nonwork. *Journal of Vocational Behavior*, 68(3), 432–445. doi:10.1016/j.jvb.2005.10.006
- Park, Y., Fritz, C., & Jex, S. M. (2011). Relationships between work-home segmentation and psychological detachment from work: The role of communication technology use at home. *Journal of Occupational Health Psychology*, 16(4), 457–467. doi:10.1037/a0023594
- Peluchette, J. V., Karl, K., & Fertig, J. (2013). A Facebook 'friend' request from the boss: Too close for comfort? *Business Horizons*, 56(3), 291–300. doi:10.1016/j.bushor.2013.01.013
- Pike, J. C., Bateman, P. J., & Butler, B. (2013). Dialectic tensions of information quality: Social networking sites and hiring. *Journal of Computer-Mediated Communication*, 19(1), 56–77. doi:10.1111/jcc4.12031

- Rothbard, N. P., Phillips, K. W., & Dumas, T. L. (2005). Managing multiple roles: Work-family policies and individuals' desires for segmentation. *Organization Science*, 16(3), 243–258. doi:10.1287/orsc.1050.0124
- Schaufeli, W. B., & Bakker, A. B. (2004). Job demands, job resources, and their relationship with burnout and engagement: A multi-sample study. *Journal of Organizational Behavior*, 25(3), 293–315. doi:10.1002/job.248
- Stohl, C., Etter, M., Banghart, S., & Woo, D. (2017). Social media policies: Implications for contemporary notions of corporate social responsibility. *Journal of Business Ethics*, 142(3), 413–436. doi:10.1007/s10551-015-2743-9
- Treem, J. W., & Leonardi, P. M. (2013). Social media use in organizations: Exploring the affordances of visibility, editability, persistence, and association. *Annals of the International Communication Association*, 36(1), 143–189. doi:10.1080/23808985.2013.11679130
- van Zoonen, W., Verhoeven, J. W. M., & Vliegenthart, R. (2016). Social media's dark side: Inducing boundary conflicts. *Journal of Managerial Psychology*, 31(8), 1–15. doi:10.1108/JMP-10-2015-0388
- van Zoonen, W., Verhoeven, J. W. M., & Vliegenthart, R. (2017). Understanding the consequences of public social media use for work. *European Management Journal*, 35(5), 595–605. doi:10.1016/j.emj.2017.07.006